Roll No. ....

TCS–391

B. TECH. (CSE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

FUNDAMENTAL OF CYBER SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the use of the zero trust model in cyber security ? Provide the details of key steps of the zero trust model. (CO1)

OR

(b) Why is it important to achieve data integrity ? Explain the procedure of protection against the data modification attack using a figure. (CO1)

2. (a) What is penetration testing ? Explain all different stages of penetration testing. (CO2)

OR

(b) Create an access matrix for the following scenario : (CO2)

There are four users (i.e., U1, U2, U3, U4) in the system.

There are four objects/files (i.e., F1, F2, F3, F4) in the system.

User U1 cannot access F1, can read F2 and F3 and read/write F4.

User U2 can read/write F1, F2 and F3 and can read/write/execute F4.

User U3 can read/write F1 and F2 and can read/write/execute F3 and F4.

User U4 can read/write/execute all files.

3. (a) Explain any *two* methods, which can be used to provide protection against unauthorized access attempts. (CO3)

OR

(b) How would you compare "denial-of-service (DoS) attack and distributed denial-of-service (DDoS) attack" ? Which one is more devastating ? (CO3)

4. (a) How would you compare "black hat hackers and grey hat hackers" ? Who are more devastating ? (CO4)

OR

(b) Write down a python program to store the details of 100 credit card holders in a file. Use the concept of python file handling. Following parameters should be stored in the created file :

Card holder name, hash value of credit card number, hash value of expiry date.

Take the values of card holder name, credit card number and expiry date as the inputs from the user. (CO4)

5. (a) Why do we need cyber security ? What are the three pillars of cyber security ? (CO6)

OR

(b) What is the use of an access control mechanism ? How would you design an access control mechanism for the smart healthcare system ? (CO6)